

Abstract algebra – 3

Permutations

Homework

13 сентября 2026 года

Theoretical minimum. Every permutation splits uniquely into disjoint cycles, which are the orbits of $\langle \sigma \rangle$; the order of σ is the least common multiple of the cycle lengths, and two permutations are conjugate in S_n exactly when they have the same cycle type. The sign satisfies $\text{sgn}(\sigma) = (-1)^{\text{inv}(\sigma)} = (-1)^{n-c(\sigma)}$, where inv counts inversions and c counts cycles including fixed points; it is a homomorphism onto $\{\pm 1\}$ with kernel A_n . Transpositions generate S_n , and already $(1\ 2)$ together with $(1\ 2\ \dots\ n)$ do; 3-cycles generate A_n . Inside linear algebra permutations appear through $\det A = \sum_{\sigma} \text{sgn}(\sigma) \prod_i a_{i\sigma(i)}$ and through the permutation matrices, for which $\det P_{\sigma} = \text{sgn}(\sigma)$ and $\text{tr } P_{\sigma}$ is the number of fixed points. For a random permutation the expected number of fixed points is 1, the expected number of cycles is $H_n \sim \ln n$, and the probability of having no fixed point tends to $1/e$.

1. Prove the following proposition for $n = 3$ and $n = 5$, and disprove it for $n = 4$: for every permutation $\pi_1 \neq \text{id}$ of $\{1, \dots, n\}$ there is a permutation π_2 such that π_1 and π_2 generate the whole of S_n . (IMC)
2. Let $n \geq 3$ and let G be a subgroup of S_n generated by $n-2$ transpositions. For $k \in \{1, \dots, n\}$ put $S(k) = \{\sigma(k) : \sigma \in G\}$. Show that $|S(k)| \leq n-1$ for every k . (Румыния, 2007)
3. Let G be a subgroup of S_n such that every $\pi \in G \setminus \{e\}$ has exactly one fixed point. Show that this fixed point is the same for all such π . (IMC)

4. For $\pi \in S_n$ let $\text{inv}(\pi)$ be the number of inversions, and let $f(n)$ be the number of $\pi \in S_n$ for which $\text{inv}(\pi)$ is divisible by $n + 1$. Prove that there are infinitely many primes p with $f(p - 1) > \frac{(p-1)!}{p}$ and infinitely many primes p with $f(p - 1) < \frac{(p-1)!}{p}$. (IMC 2016)
5. Find all $n \in \mathbb{N}$ for which there exist two cycles c_1, c_2 of length n in S_n that together generate S_n . (ISTCiM 2024)
6. Let $a_1, \dots, a_n$ be a non-constant arithmetic progression of real numbers, and form an $n \times n$ matrix whose rows are n pairwise different cyclic permutations of $a_1, \dots, a_n$. Prove that its determinant vanishes if and only if $a_1 + a_n = 0$. (A related counting problem about choosing one entry in each row and column is Linear algebra – 6, homework 5.) (IMC 2025, shortlist)
7. Let P_n be the number of permutations π of $\{1, \dots, n\}$ such that $|i - j| = 1$ implies $|\pi(i) - \pi(j)| \leq 2$. Show that for $n \geq 2$ the quantity $P_{n+5} - P_{n+4} - P_{n+3} + P_n$ does not depend on n , and find its value. (Putnam 2015, B5)
8. Let $p > 2$ be a prime. Prove that there is a permutation $(x_1, x_2, \dots, x_{p-1})$ of the numbers $(1, 2, \dots, p-1)$ such that $x_1x_2 + x_2x_3 + \dots + x_{p-2}x_{p-1} \equiv 2 \pmod{p}$. (IMC)
9. For $\sigma = (i_1, \dots, i_n) \in S_n$ put $D(\sigma) = \sum_{k=1}^n |i_k - k|$, and let $Q(n, d)$ be the number of permutations with $D(\sigma) = d$. Prove that $Q(n, d)$ is even for $d \geq 2n$. *Hint:* consider the determinant of the matrix with entries $x^{|i-j|}$. (IMC)
10. Prove that the permanent of the $n \times n$ matrix of ones minus the identity equals the number of permutations without fixed points, and use the expansion of the determinant to prove the inclusion–exclusion formula for that number. Then show that the permutation matrices span the space of all matrices with equal row and column sums.